

Exploratory analysis of a public 16S rRNA microbiome dataset

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0.1 Readme & Timeline

This project conducts an exploratory downstream analysis of processed 16S rRNA amplicon sequencing data from the GlobalPatterns dataset. It evaluates sequencing depth, alpha diversity, beta diversity, taxonomic composition, and differences in microbial community structure across sample types using R.

0.2 Settings

0.3 Required packages

- **tidyverse**: Provides tools for importing, cleaning, transforming, summarising, and visualising tabular data, including `dplyr`, `readr`, `tidyr`, and `ggplot2`.
- **phyloseq**: Integrates microbiome abundance data, taxonomy, phylogenetic information, and sample metadata into a single object for ecological analysis.
- **vegan**: Provides ecological statistics, including Bray–Curtis dissimilarity calculations, ordination support, and PERMANOVA.
- **microbiome**: Provides microbiome-specific functions for abundance transformation, prevalence filtering, and community composition analysis.
- **ape**: Provides functions for reading, handling, and analysing phylogenetic trees when a tree is included in the dataset.
- **janitor**: Provides convenient functions for cleaning and standardising variable and column names.
- **here**: Creates reliable relative file paths within the project folder, improving reproducibility across computers.
- **rmarkdown**: Renders the analysis, code, figures, and written interpretation into a reproducible HTML report.
- **knitr**: Executes R code chunks and formats code, tables, figures, and output inside the R Markdown report.
- **tinytex**: A lightweight, cross-platform, portable, and easy-to-maintain LaTeX distribution based on TeX Live

1 Introduction and Research Question

Research question:

How do microbial alpha diversity, beta diversity, and taxonomic composition vary across environmental and host-associated sample types in the GlobalPatterns dataset?

Microbial communities differ substantially depending on the environment or host they are associated with. This analysis explores whether samples from different environmental and host-associated sources (e.g., soil, ocean, feces, skin) differ in:

- **Alpha diversity** — diversity *within* a sample (richness/evenness)
- **Beta diversity** — diversity *between* samples (community structure)
- **Taxonomic composition** — which taxa dominate each sample type

2 Dataset Description

GlobalPatterns is a widely used example 16S rRNA amplicon dataset distributed with the `phyloseq` package (Caporaso et al. 2011). It contains high-throughput sequencing data from 26 samples spanning diverse environmental and host-associated sources, along with an associated phylogenetic tree and taxonomy assignments.

3 Package Setup

```
library(tidyverse)
library(vegan)
library(ape)
library(rmarkdown)
library(knitr)
library(here)
library(janitor)
library(phyloseq)
library(microbiome)
```

4 Load GlobalPatterns

```
data(GlobalPatterns)
GlobalPatterns

## phyloseq-class experiment-level object
## otu_table() OTU Table: [ 19216 taxa and 26 samples ]
## sample_data() Sample Data: [ 26 samples by 7 sample variables ]
## tax_table() Taxonomy Table: [ 19216 taxa by 7 taxonomic ranks ]
## phy_tree() Phylogenetic Tree: [ 19216 tips and 19215 internal nodes ]
```

5 Inspect Data Structure

```
ntaxa(GlobalPatterns)
```

```
## [1] 19216
```

```
nsamples(GlobalPatterns)
```

```
## [1] 26
```

```
rank_names(GlobalPatterns)
```

```
## [1] "Kingdom" "Phylum" "Class" "Order" "Family" "Genus" "Species"
```

```
# Zero-count taxa (candidates for filtering later)
sum(taxa_sums(GlobalPatterns) == 0)
```

```
## [1] 228
```

```
# Any missing values in the OTU table?
any(is.na(otu_table(GlobalPatterns)))
```

```
## [1] FALSE
```

6 Inspect Sample Metadata

```
sample_data(GlobalPatterns) %>%
  as("data.frame") %>%
  glimpse()
```

```
## Rows: 26
## Columns: 7
## $ X.SampleID      <fct> CL3, CC1, SV1, M31Fcsw, M11Fcsw, M31Plmr, M11~
## $ Primer          <fct> ILBC_01, ILBC_02, ILBC_03, ILBC_04, ILBC_05, ~
## $ Final_Barcode    <fct> AACGCA, AACTCG, AACTGT, AAGAGA, AAGCTG, AATCG~
## $ Barcode_truncated_plus_T <fct> TGCGTT, CGAGTT, ACAGTT, TCTCTT, CAGCTT, ACAT~
## $ Barcode_full_length <fct> CTAGCGTGCGT, CATCGACGAGT, GTACGCACAGT, TCGACA~
## $ SampleType       <fct> Soil, Soil, Soil, Feces, Feces, Skin, Skin, S~
## $ Description      <fct> "Calhoun South Carolina Pine soil, pH 4.9", "~
```

```
# Sample types available for grouping in later analyses
sample_data(GlobalPatterns)$SampleType %>% unique()
```

```
## [1] Soil      Feces      Skin      Tongue
## [5] Freshwater Freshwater (creek) Ocean      Sediment (estuary)
## [9] Mock
## 9 Levels: Feces Freshwater Freshwater (creek) Mock ... Tongue
```

```
# Sample count per SampleType
sample_data(GlobalPatterns) %>%
  as("data.frame") %>%
  count(SampleType, sort = TRUE)
```

```
##           SampleType n
## 1           Feces 4
## 2 Freshwater (creek) 3
## 3           Mock 3
## 4           Ocean 3
## 5 Sediment (estuary) 3
## 6           Skin 3
## 7           Soil 3
## 8      Freshwater 2
## 9           Tongue 2
```

7 Sequencing-Depth Assessment

Sequencing depth (library size = total reads per sample) varies naturally across samples and sample types. Low-depth samples can artificially deflate alpha diversity estimates and distort beta diversity distances, so we inspect depth here before deciding on any filtering or exclusion.

```
# Build a depth table: one row per sample, with SampleType attached
depth_df <- tibble(
  SampleID = sample_names(GlobalPatterns),
  LibSize  = sample_sums(GlobalPatterns),
  SampleType = sample_data(GlobalPatterns)$SampleType
)

depth_df %>% arrange(LibSize)
```

```
## # A tibble: 26 x 3
##   SampleID LibSize SampleType
##   <chr>      <dbl> <fct>
## 1 TRRsed1    58688 Sediment (estuary)
## 2 M11Tong   100187 Tongue
## 3 F21Plmr   186297 Skin
## 4 TRRsed3   279704 Sediment (estuary)
## 5 M11Plmr   433894 Skin
## 6 TRRsed2   493126 Sediment (estuary)
## 7 NP2       523634 Ocean
## 8 SV1       697509 Soil
## 9 M31Plmr   718943 Skin
## 10 CL3      864077 Soil
## # i 16 more rows
```

```
# Numeric summary overall
summary(depth_df$LibSize)
```

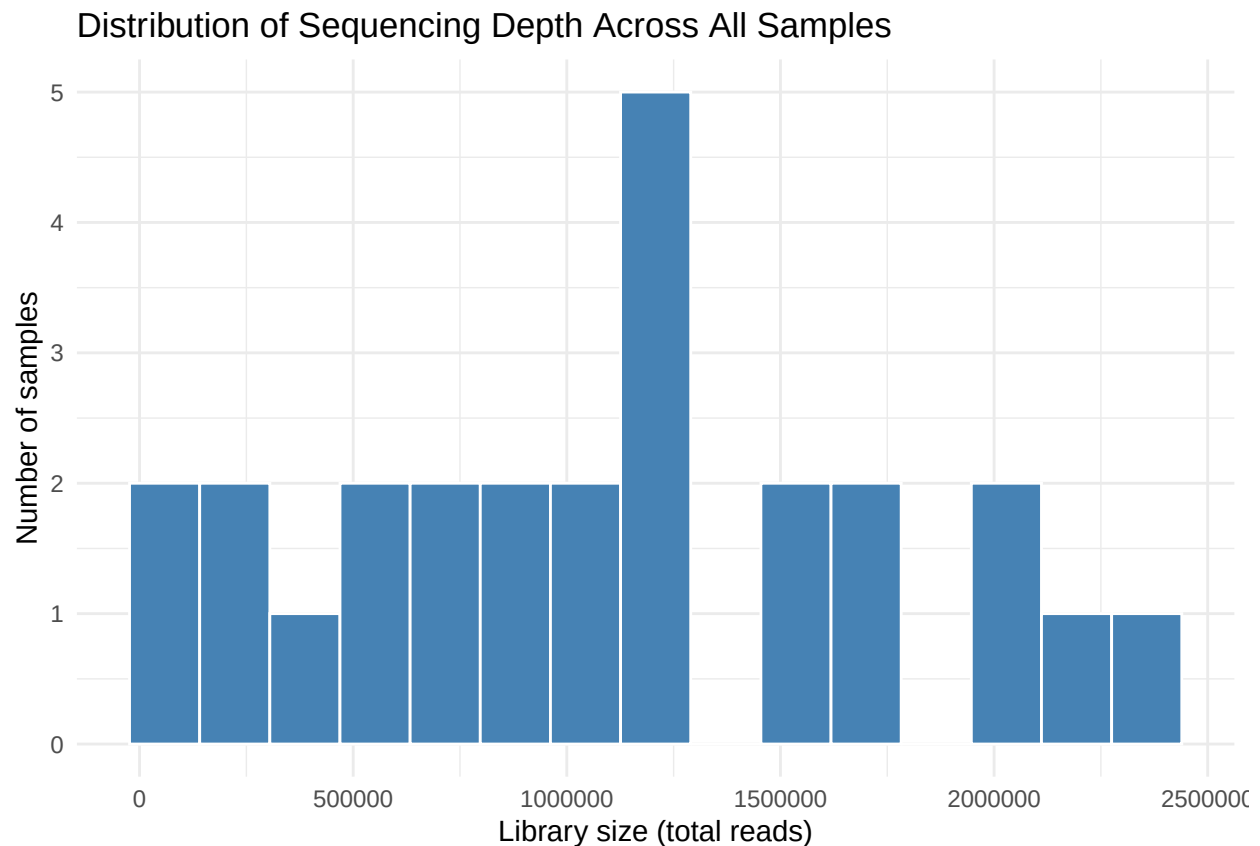
```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  58688  567103 1106849 1085257 1527330 2357181
```

```
# Numeric summary by SampleType
depth_df %>%
  group_by(SampleType) %>%
  summarise(
    n      = n(),
    min    = min(LibSize),
    median = median(LibSize),
    mean   = mean(LibSize),
    max    = max(LibSize),
    .groups = "drop"
  ) %>%
  arrange(median)
```

```
## # A tibble: 9 x 6
##   SampleType      n    min  median  mean  max
##   <fct>      <int> <dbl>   <dbl> <dbl> <dbl>
```

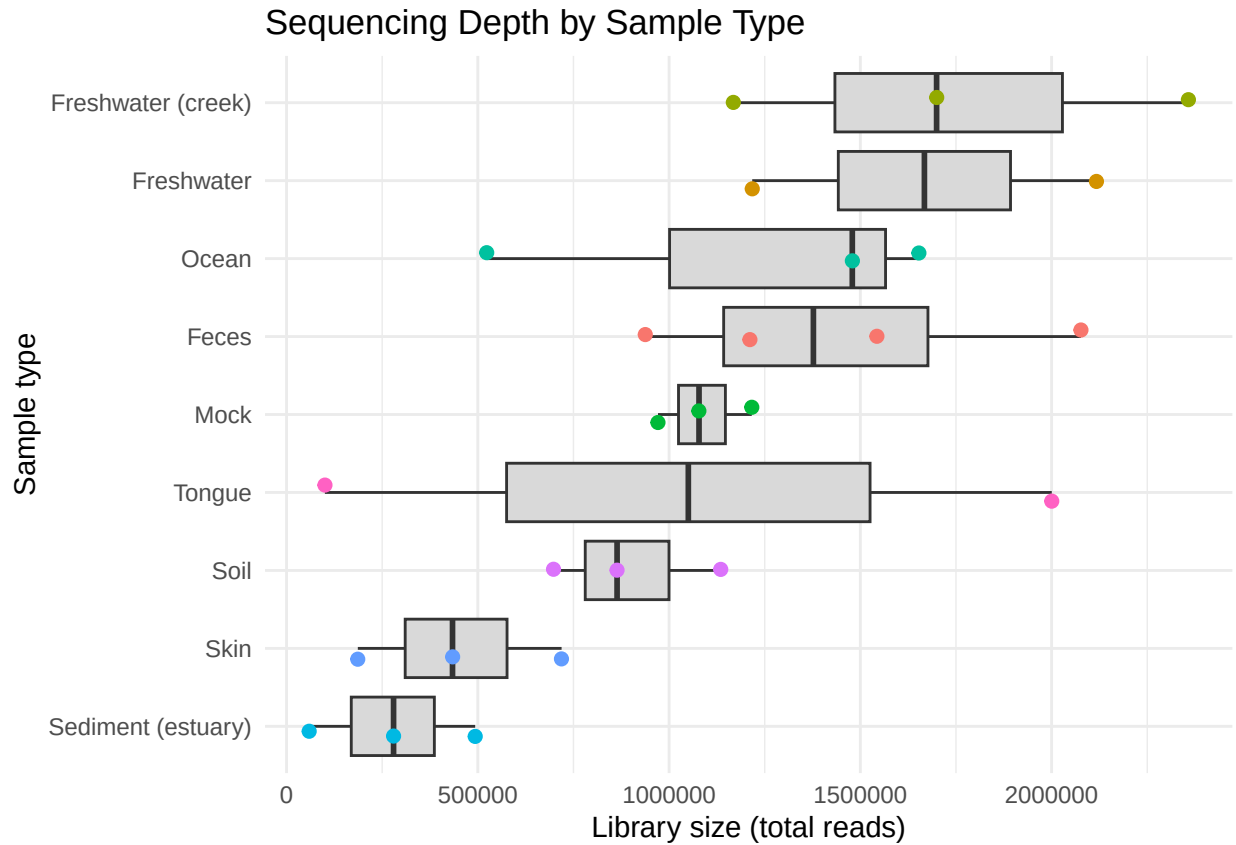
## 1 Sediment (estuary)	3	58688	279704	277173.	493126
## 2 Skin	3	186297	433894	446378	718943
## 3 Soil	3	697509	864077	899014.	1135457
## 4 Tongue	2	100187	1050294.	1050294.	2000402
## 5 Mock	3	971073	1078241	1088484.	1216137
## 6 Feces	4	937466	1377261	1442116	2076476
## 7 Ocean	3	523634	1478965	1218451	1652754
## 8 Freshwater	2	1217312	1667452	1667452	2117592
## 9 Freshwater (creek)	3	1167748	1699293	1741407.	2357181

```
ggplot(depth_df, aes(x = LibSize)) +
  geom_histogram(bins = 15, fill = "steelblue", color = "white") +
  labs(
    title = "Distribution of Sequencing Depth Across All Samples",
    x = "Library size (total reads)",
    y = "Number of samples"
  ) +
  theme_minimal()
```



```
ggplot(depth_df, aes(x = reorder(SampleType, LibSize, median), y = LibSize)) +
  geom_boxplot(outlier.shape = NA, fill = "grey85") +
  geom_jitter(aes(color = SampleType), width = 0.15, size = 2, show.legend = FALSE) +
  coord_flip() +
  labs(
    title = "Sequencing Depth by Sample Type",
  )
```

```
x = "Sample type",
y = "Library size (total reads)"
) +
theme_minimal()
```



```
# Flag samples below the 10th percentile of overall library size as a
# starting heuristic; adjust threshold after visual inspection above.
low_depth_threshold <- quantile(depth_df$LibSize, 0.10)
```

```
depth_df %>%
  filter(LibSize < low_depth_threshold) %>%
  arrange(LibSize)
```

```
## # A tibble: 3 x 3
##   SampleID LibSize SampleType
##   <chr>      <dbl> <fct>
## 1 TRRsed1    58688 Sediment (estuary)
## 2 M11Tong   100187 Tongue
## 3 F21Plmr   186297 Skin
```

```
# Check for natural gaps / isolated low-depth failures by looking at the
# ratio between each sample and the next-lowest one, once sorted by depth.
```

```
depth_df %>%
  arrange(LibSize) %>%
```

```
mutate(
  PreviousDepth = lag(LibSize),
  RatioToPrevious = LibSize / PreviousDepth
)
```

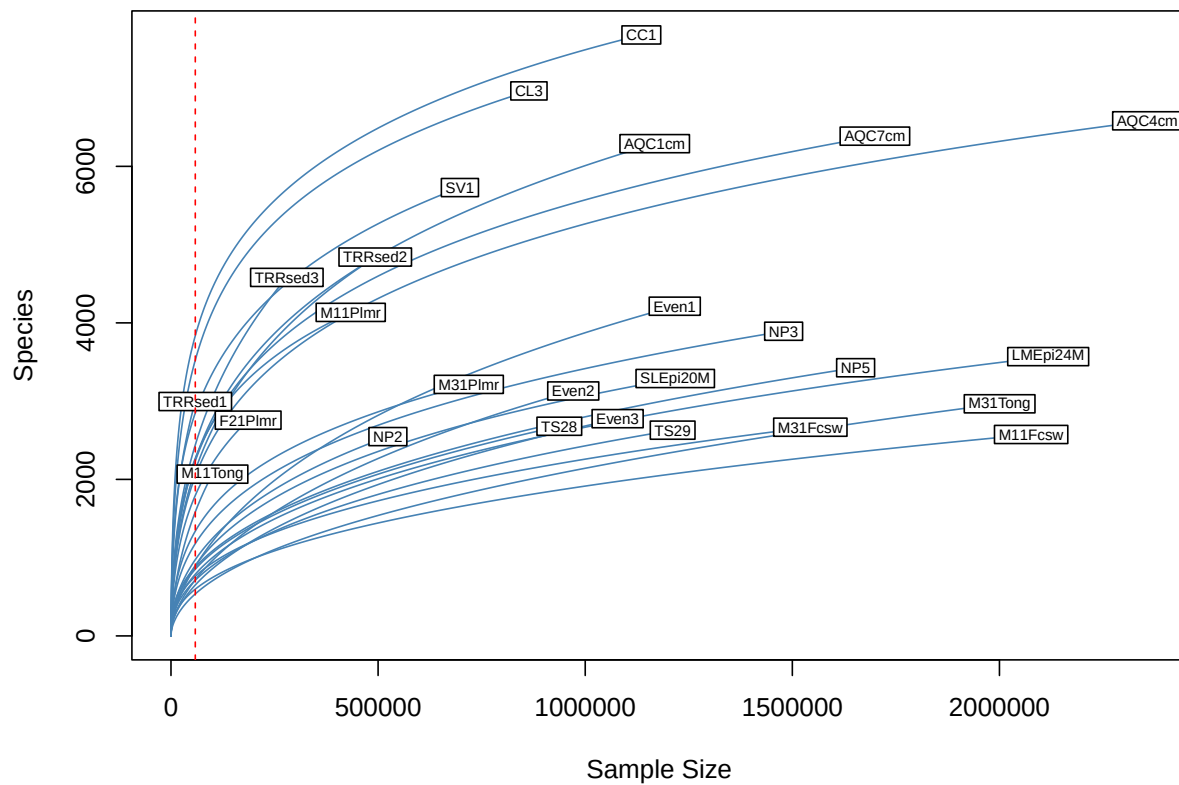
```
## # A tibble: 26 x 5
##   SampleID LibSize SampleType PreviousDepth RatioToPrevious
##   <chr>      <dbl> <fct>          <dbl>          <dbl>
## 1 TRRsed1    58688 Sediment (estuary)      NA            NA
## 2 M11Tong   100187 Tongue              58688          1.71
## 3 F21Plmr   186297 Skin                100187          1.86
## 4 TRRsed3   279704 Sediment (estuary)    186297          1.50
## 5 M11Plmr   433894 Skin                279704          1.55
## 6 TRRsed2   493126 Sediment (estuary)    433894          1.14
## 7 NP2       523634 Ocean                493126          1.06
## 8 SV1       697509 Soil                 523634          1.33
## 9 M31Plmr   718943 Skin                 697509          1.03
## 10 CL3      864077 Soil                 718943          1.20
## # i 16 more rows
```

*# Rarefaction curves: if a sample's curve plateaus well before its total
depth, its diversity estimate is not depth-limited. If curves for
low-depth samples end while still rising steeply, that IS a depth-driven
failure signal (unlike a simple ratio/threshold check).*

```
otu_mat <- as(otu_table(GlobalPatterns), "matrix")
if (taxa_are_rows(GlobalPatterns)) otu_mat <- t(otu_mat)
```

```
rarecurve(
  otu_mat,
  step = 1000,
  label = TRUE,
  cex = 0.6,
  col = "steelblue",
  main = "Rarefaction Curves by Sample"
)
abline(v = min(depth_df$LibSize), col = "red", lty = 2)
```

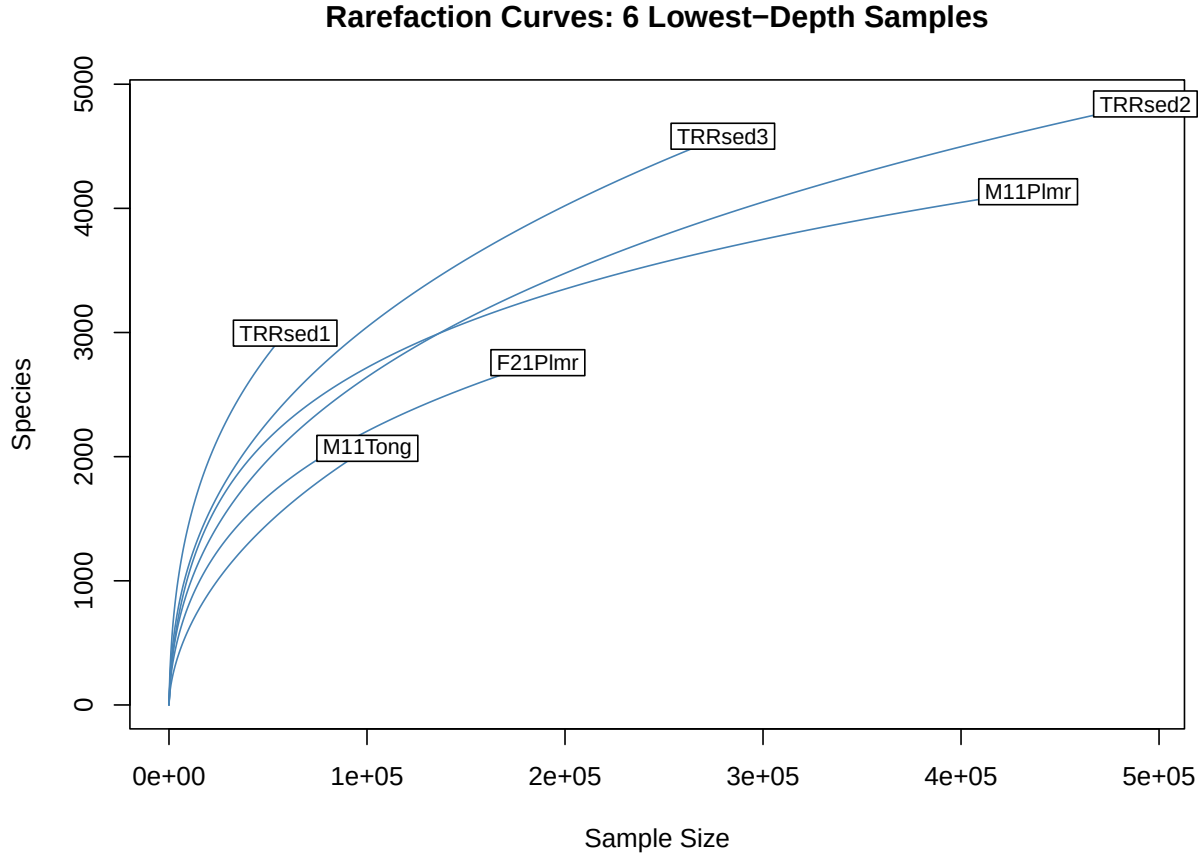
Rarefaction Curves by Sample



*# Zoom in on the lowest-depth samples only, so the slope near each curve's
endpoint can actually be judged (the full 26-sample plot is too compressed
to read this reliably).*

```
low_depth_samples <- depth_df %>%
  arrange(LibSize) %>%
  slice_head(n = 6) %>%
  pull(SampleID)

rarecurve(
  otu_mat[low_depth_samples, ],
  step = 500,
  label = TRUE,
  cex = 0.8,
  col = "steelblue",
  main = "Rarefaction Curves: 6 Lowest-Depth Samples"
)
```



Observed result: The dashed red line marks the minimum observed library size (58,688 reads; TRRsed1). Curves for the lowest-depth samples were still increasing near their endpoints, suggesting that sequencing did not fully capture all rare features. Alpha-diversity estimates should therefore be interpreted as conservative lower-bound estimates of richness.

8 Filtering Decision

8.0.1 Decision on low-depth samples

All 26 samples were retained. Although three samples fell below the 10th percentile of sequencing depth, none showed a clear isolated low-depth failure, and excluding samples would further reduce already small group sizes. Alpha diversity was calculated from rarefied counts at the minimum observed depth, while beta diversity used relative-abundance profiles. Rarefaction curves for the lowest-depth samples were still increasing at their endpoints; richness estimates should therefore be interpreted as conservative lower bounds. Features with zero counts across all samples were removed before downstream analysis.

8.0.2 Taxon-level filtering

Separately from sample filtering, taxa with zero counts across all samples add no information and are removed before downstream analysis.

```
# Remove taxa with zero total counts across all samples
GlobalPatterns_filt <- prune_taxa(taxa_sums(GlobalPatterns) > 0, GlobalPatterns)
```

```
ntaxa(GlobalPatterns)      # before
```

```
## [1] 19216
```

```
ntaxa(GlobalPatterns_filt) # after
```

```
## [1] 18988
```

```
before <- ntaxa(GlobalPatterns)
after  <- ntaxa(GlobalPatterns_filt)

percent_removed <- (before - after) / before * 100
percent_retained <- after / before * 100
percent_removed
```

```
## [1] 1.186511
```

```
percent_retained
```

```
## [1] 98.81349
```

8.0.3 Prepare normalized objects for downstream analysis

```
set.seed(123) # reproducibility for rarefaction

# 1. Rarefied object -> for alpha diversity (section 9)
#   Rarefies every sample to the minimum observed depth so richness/
#   Shannon are compared on an equal footing.
GlobalPatterns_rare <- rarefy_even_depth(
  GlobalPatterns_filt,
  sample.size = min(sample_sums(GlobalPatterns_filt)),
  rngseed     = 123,
  replace     = FALSE,
  verbose     = FALSE
)

ntaxa(GlobalPatterns_rare)
```

```
## [1] 12840
```

```
sample_sums(GlobalPatterns_rare) %>% unique() # should all equal min depth
```

```
## [1] 58688
```

```

# 2. Relative-abundance object -> for beta diversity / PCoA (section 10)
#   Converts counts to proportions per sample, avoiding the need to
#   discard reads via rarefaction for ordination-based comparisons.
GlobalPatterns_relabund <- transform_sample_counts(
  GlobalPatterns_filt,
  function(x) x / sum(x)
)

sample_sums(GlobalPatterns_relabund) %>% unique() # should all equal 1

```

```
## [1] 1
```

9 Alpha diversity

How diverse is each individual microbiome sample?

Alpha diversity describes the diversity within an individual microbiome sample. This analysis uses the rarefied feature table, in which all samples contain the same number of feature counts, to reduce the influence of unequal sequencing depth.

Two metrics are examined:

- **Observed richness:** the number of observed microbial features in a sample.
- **Shannon diversity:** a measure that reflects both the number of features and the evenness of their relative abundances.

```

alpha_diversity <- estimate_richness(
  GlobalPatterns_rare,
  measures = c("Observed", "Shannon", "Simpson")
) |>
  tibble::rownames_to_column("SampleID") |>
  left_join(
    depth_df |>
      select(SampleID, SampleType, LibSize),
    by = "SampleID"
  )

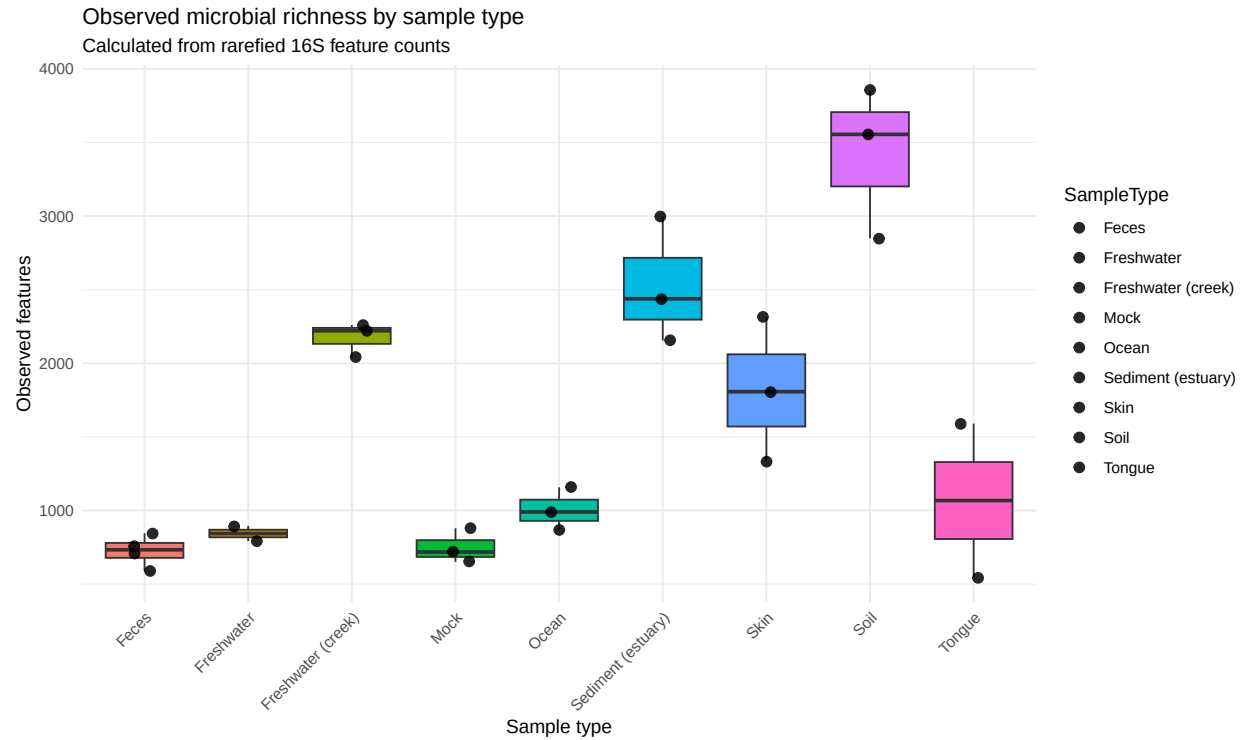
```

```
alpha_diversity
```

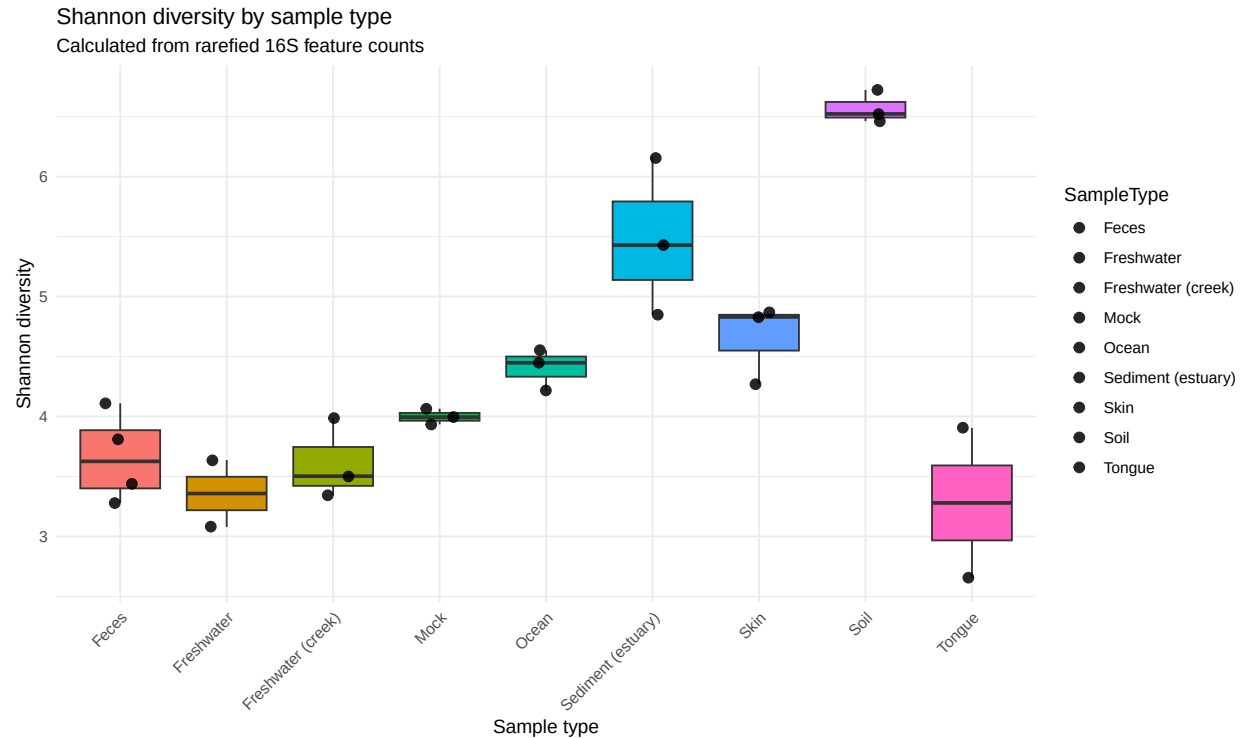
	SampleID	Observed	Shannon	Simpson	SampleType	LibSize
## 1	CL3	3555	6.523026	0.9945306	Soil	864077
## 2	CC1	3857	6.723098	0.9951734	Soil	1135457
## 3	SV1	2848	6.462136	0.9962425	Soil	697509
## 4	M31Fcsw	758	3.811084	0.9270416	Feces	1543451
## 5	M11Fcsw	589	3.279804	0.9106837	Feces	2076476
## 6	M31Plmr	1334	4.269331	0.9379325	Skin	718943
## 7	M11Plmr	2315	4.830150	0.9519636	Skin	433894
## 8	F21Plmr	1807	4.866082	0.9779074	Skin	186297
## 9	M31Tong	545	2.654647	0.8616103	Tongue	2000402
## 10	M11Tong	1590	3.904547	0.9364362	Tongue	100187
## 11	LMEpi24M	792	3.079271	0.8028816	Freshwater	2117592

## 12	SLEpi20M	895	3.636902	0.9069612	Freshwater	1217312
## 13	AQC1cm	2260	3.502747	0.7632103	Freshwater (creek)	1167748
## 14	AQC4cm	2043	3.341037	0.7386072	Freshwater (creek)	2357181
## 15	AQC7cm	2221	3.988752	0.8178390	Freshwater (creek)	1699293
## 16	NP2	990	4.216211	0.9531651	Ocean	523634
## 17	NP3	1157	4.447797	0.9716424	Ocean	1478965
## 18	NP5	869	4.552264	0.9747965	Ocean	1652754
## 19	TRRsed1	2995	6.157462	0.9924388	Sediment (estuary)	58688
## 20	TRRsed2	2155	4.848279	0.9645751	Sediment (estuary)	493126
## 21	TRRsed3	2438	5.428607	0.9816982	Sediment (estuary)	279704
## 22	TS28	845	4.110183	0.9653290	Feces	937466
## 23	TS29	708	3.441169	0.9185227	Feces	1211071
## 24	Even1	879	4.065167	0.9682462	Mock	1216137
## 25	Even2	718	3.934744	0.9635913	Mock	971073
## 26	Even3	651	3.995106	0.9672839	Mock	1078241

```
ggplot(
  alpha_diversity,
  aes(x = SampleType, y = Observed, fill = SampleType)
) +
  geom_boxplot(show.legend = FALSE, outlier.shape = NA) +
  geom_jitter(width = 0.15, size = 2.5, alpha = 0.85) +
  labs(
    title = "Observed microbial richness by sample type",
    subtitle = "Calculated from rarefied 16S feature counts",
    x = "Sample type",
    y = "Observed features"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```



```
ggplot(
  alpha_diversity,
  aes(x = SampleType, y = Shannon, fill = SampleType)
) +
  geom_boxplot(show.legend = FALSE, outlier.shape = NA) +
  geom_jitter(width = 0.15, size = 2.5, alpha = 0.85) +
  labs(
    title = "Shannon diversity by sample type",
    subtitle = "Calculated from rarefied 16S feature counts",
    x = "Sample type",
    y = "Shannon diversity"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
)
```



```
alpha_summary <- alpha_diversity |>
  group_by(SampleType) |>
  summarise(
    Samples = n(),
    MedianObserved = median(Observed),
    MedianShannon = median(Shannon),
    MedianSimpson = median(Simpson),
    .groups = "drop"
  ) |>
  arrange(desc(MedianShannon))
```

alpha_summary

```
## # A tibble: 9 x 5
##   SampleType      Samples MedianObserved MedianShannon MedianSimpson
##   <fct>          <int>         <dbl>         <dbl>         <dbl>
## 1 Soil              3          3555           6.52           0.995
## 2 Sediment (estuary) 3          2438           5.43           0.982
## 3 Skin              3          1807           4.83           0.952
## 4 Ocean             3           990           4.45           0.972
## 5 Mock              3           718           4.00           0.967
## 6 Feces             4           733           3.63           0.923
## 7 Freshwater (creek) 3          2221           3.50           0.763
## 8 Freshwater        2           844           3.36           0.855
## 9 Tongue            2          1068           3.28           0.899
```

Observed result: Soil samples showed the highest richness and Shannon diversity, indicating diverse and relatively even communities. Freshwater creek samples had high richness but low Shannon diversity, suggesting uneven feature abundances. Results are descriptive because groups were small and heterogeneous.

9.0.1 Statistical test

Kruskal-Wallis tests assessed overall differences across sample types because group sizes were small and unequal ($n = 2-4$) and normality could not be assumed.

```
kruskal.test(Observed ~ SampleType, data = alpha_diversity)

##
## Kruskal-Wallis rank sum test
##
## data: Observed by SampleType
## Kruskal-Wallis chi-squared = 21.33, df = 8, p-value = 0.006319
```

```
kruskal.test(Shannon ~ SampleType, data = alpha_diversity)

##
## Kruskal-Wallis rank sum test
##
## data: Shannon by SampleType
## Kruskal-Wallis chi-squared = 22.125, df = 8, p-value = 0.004689
```

```
kruskal.test(Simpson ~ SampleType, data = alpha_diversity)

##
## Kruskal-Wallis rank sum test
##
## data: Simpson by SampleType
## Kruskal-Wallis chi-squared = 21.416, df = 8, p-value = 0.006121
```

Result: Alpha diversity differed across sample types for all three metrics: Observed richness ($\chi^2 = 21.33$, $df = 8$, $p = 0.0063$), Shannon diversity ($\chi^2 = 22.13$, $df = 8$, $p = 0.0047$), and Simpson diversity ($\chi^2 = 21.42$, $df = 8$, $p = 0.0061$). This supports the descriptive pattern of higher diversity in soil and lower diversity in some host-associated or mock samples.

The significant omnibus tests indicate that at least one sample type differs from another, but do not identify specific group pairs. Because groups contain only 2–4 samples, results are exploratory and no pairwise post-hoc comparisons were conducted.

10 Beta diversity / PCoA

Beta diversity assesses differences in microbial community composition between samples. Bray-Curtis dissimilarity was calculated from relative-abundance profiles to account for unequal sequencing depth while retaining all observed taxa. PCoA was used to visualise the resulting between-sample relationships.

```
bray_dist <- phyloseq::distance(GlobalPatterns_relabund, method = "bray")

# Quick check: distance matrix dimensions should match sample count
as.matrix(bray_dist) %>% dim()
```

```
## [1] 26 26
```

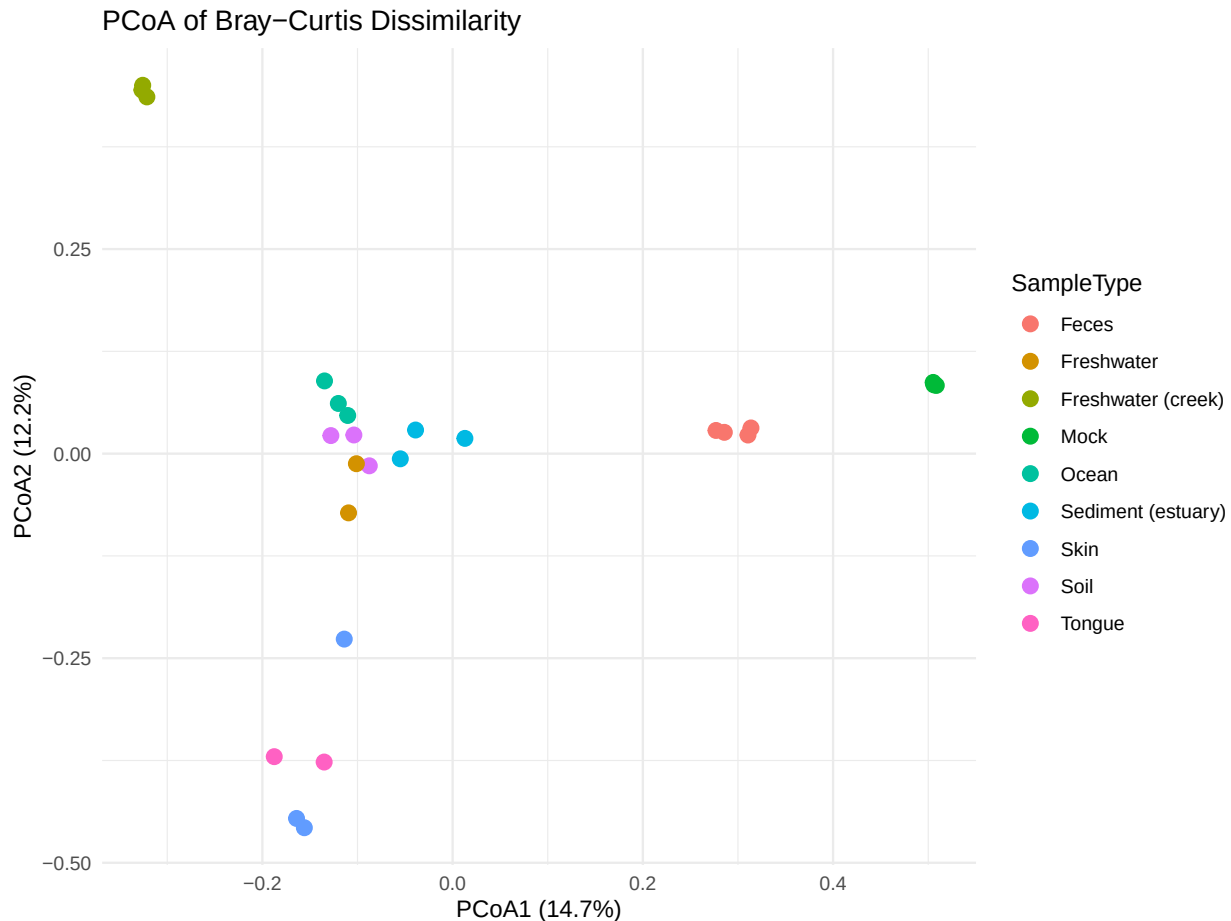
```
pcoa_ord <- ordinate(GlobalPatterns_relabund, method = "PCoA", distance = bray_dist)

# Variance explained by each axis
pcoa_eig <- pcoa_ord$values$Relative_eig
round(pcoa_eig[1:5] * 100, 1) # % variance explained, axes 1-5
```

```
## [1] 14.7 12.2 10.6 9.0 8.7
```

```
plot_ordination(GlobalPatterns_relabund, pcoa_ord, color = "SampleType") +
  geom_point(size = 3) +
  labs(
    title = "PCoA of Bray-Curtis Dissimilarity",
    x = paste0("PCoA1 (", round(pcoa_eig[1] * 100, 1), "%)"),
    y = paste0("PCoA2 (", round(pcoa_eig[2] * 100, 1), "%)"),
  ) +
  theme_minimal()
```

```
## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with 'aes()'.
## i See also 'vignette("ggplot2-in-packages")' for more information.
## i The deprecated feature was likely used in the phyloseq package.
## Please report the issue at <https://github.com/joey711/phyloseq/issues>.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



Observed result: Samples positioned closer together in the PCoA plot have more similar microbial compositions, while more distant samples have more dissimilar relative-abundance profiles.

Result: The first two PCoA axes explained 14.7% and 12.2% of Bray–Curtis compositional variation, respectively (26.9% combined). The two-dimensional plot provides a useful visual summary, although most community variation is distributed across additional dimensions.

11 PERMANOVA

PCoA visualises compositional relationships but does not test whether differences among sample types are statistically significant. PERMANOVA (`adonis2()`) tests whether microbial community composition is associated with `SampleType` by comparing within-group and between-group Bray–Curtis dissimilarities using permutations.

```
set.seed(123) # reproducibility for permutation test

sample_df <- sample_data(GlobalPatterns_relabund) %>% as("data.frame")

permanova_result <- adonis2(
  bray_dist ~ SampleType,
  data      = sample_df,
  permutations = 999
)
```

```
permanova_result
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = bray_dist ~ SampleType, data = sample_df, permutations = 999)
##           Df SumOfSqs      R2      F Pr(>F)
## Model      8   8.2466 0.73109 5.7773 0.001 ***
## Residual  17   3.0332 0.26891
## Total     25  11.2798 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Observed result:: the R2 column shows the proportion of total community-composition variance explained by SampleType; the p-value column tests whether this is greater than expected by chance, based on 999 random permutations of group labels.

11.0.1 Checking the PERMANOVA assumption

PERMANOVA assumes similar within-group dispersion (spread) across groups; if dispersion differs substantially, a significant PERMANOVA result could reflect differences in spread rather than differences in group centroids (mean composition). This is checked with `betadisper()`.

```
dispersion <- betadisper(bray_dist, sample_df$SampleType)
anova(dispersion)
```

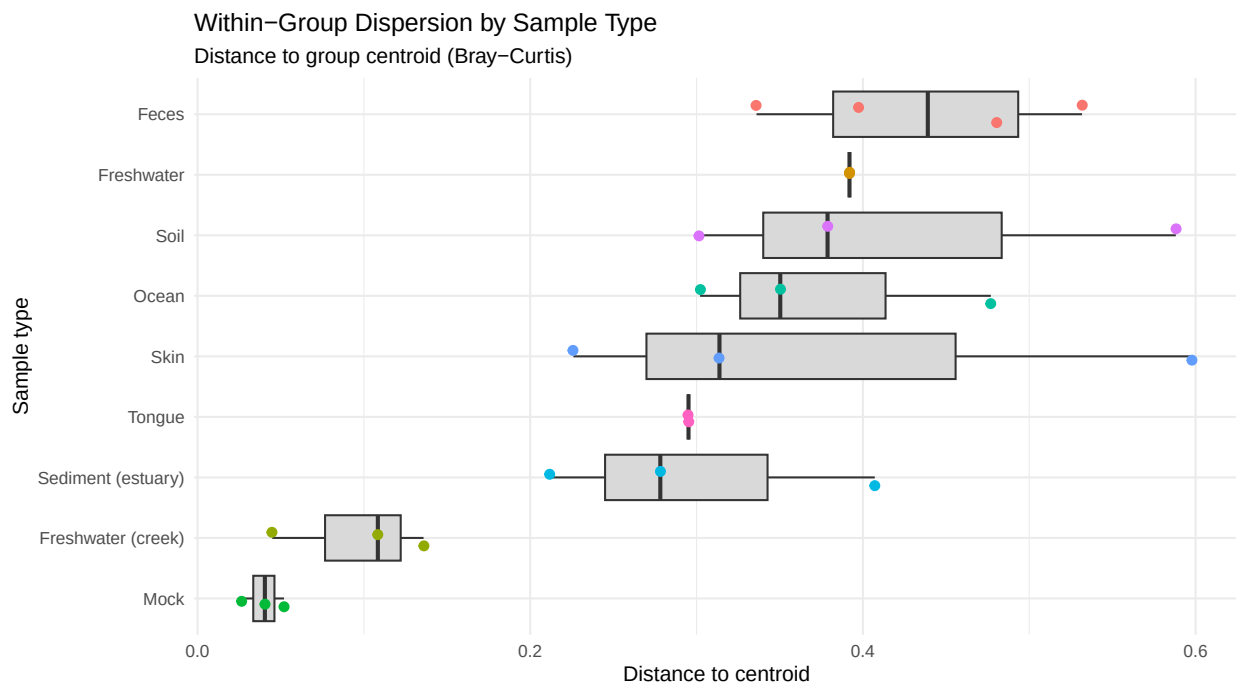
```
## Analysis of Variance Table
##
## Response: Distances
##           Df Sum Sq Mean Sq F value Pr(>F)
## Groups      8 0.49965 0.062457  5.8071 0.001158 **
## Residuals  17 0.18284 0.010755
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

PERMANOVA indicated that microbial community composition was strongly associated with SampleType ($R^2 = 0.731$, $F = 5.78$, $p = 0.001$). However, multivariate dispersion also differed among sample types ($F = 5.81$, $p = 0.0012$), indicating unequal within-group variability.

Therefore, the PERMANOVA result should be interpreted cautiously: observed separation may reflect both differences in average community composition and differences in within-group heterogeneity. Some sample types formed tight clusters of compositionally similar replicates, whereas others showed greater internal variation. Given the small and unequal group sizes ($n = 2-4$), results are exploratory rather than confirmatory.

```
dispersion_df <- tibble(
  SampleID = names(dispersion$distances),
  Distance = dispersion$distances,
  SampleType = sample_df$SampleType
)
```

```
ggplot(dispersion_df, aes(x = reorder(SampleType, Distance, median), y = Distance)) +
  geom_boxplot(fill = "grey85", outlier.shape = NA) +
  geom_jitter(aes(color = SampleType), width = 0.15, size = 2, show.legend = FALSE) +
  coord_flip() +
  labs(
    title = "Within-Group Dispersion by Sample Type",
    subtitle = "Distance to group centroid (Bray-Curtis)",
    x = "Sample type",
    y = "Distance to centroid"
  ) +
  theme_minimal()
```



This plot identifies which groups are driving the significant dispersion result. With only 2–4 points per group, the boxplots themselves are hard to read reliably — a numeric summary is more informative here.

Observed Result: Small distance → samples within that group are similar to each other → tight group / low within-group heterogeneity
 Large distance → samples within that group vary more from one another → spread-out group / high within-group heterogeneity

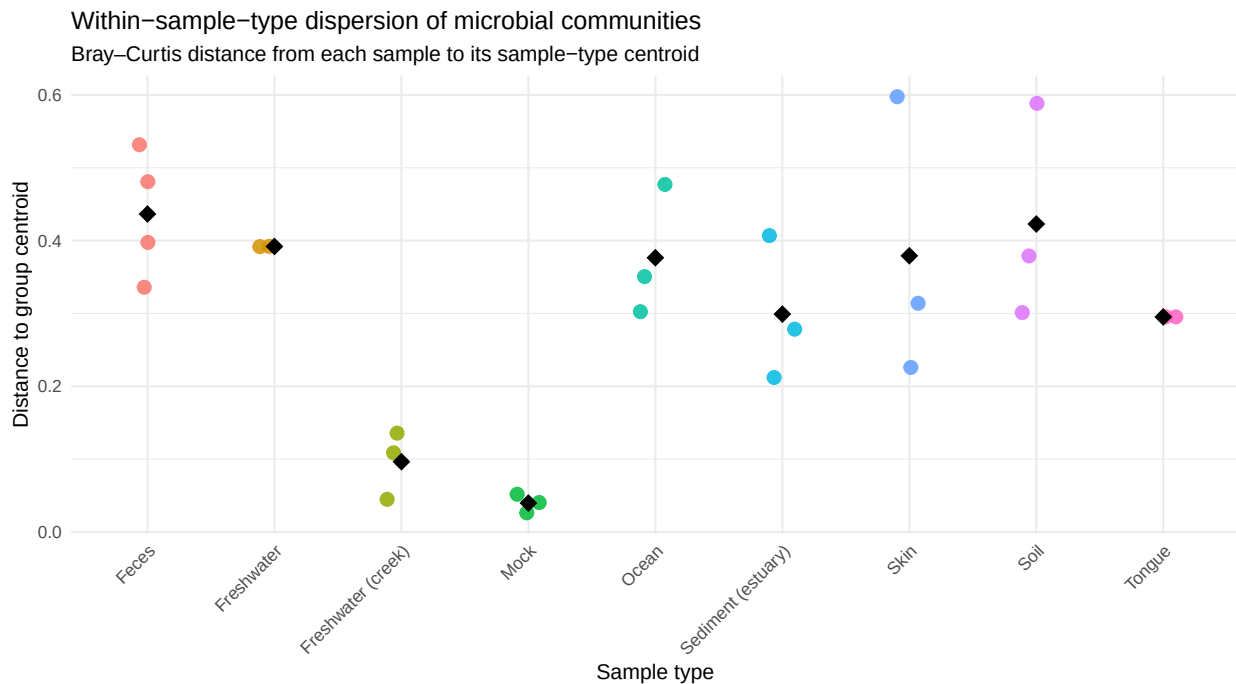
```
dispersion_df <- data.frame(
  SampleType = sample_df$SampleType,
  DistanceToCentroid = dispersion$distances
)

ggplot(
  dispersion_df,
  aes(x = SampleType, y = DistanceToCentroid, colour = SampleType)
) +
  geom_jitter(width = 0.12, size = 3, alpha = 0.85, show.legend = FALSE) +
  stat_summary(
    fun = mean,
```

```

geom = "point",
shape = 18,
size = 4,
colour = "black"
) +
labs(
  title = "Within-sample-type dispersion of microbial communities",
  subtitle = "Bray-Curtis distance from each sample to its sample-type centroid",
  x = "Sample type",
  y = "Distance to group centroid"
) +
theme_minimal() +
theme(
  axis.text.x = element_text(angle = 45, hjust = 1)
)

```



```

dispersion_df %>%
  group_by(SampleType) %>%
  summarise(
    n = n(),
    MedianDistance = median(DistanceToCentroid),
    MinDistance = min(DistanceToCentroid),
    MaxDistance = max(DistanceToCentroid),
    Range = MaxDistance - MinDistance,
    .groups = "drop"
  ) %>%
  arrange(desc(MedianDistance))

```

```
## # A tibble: 9 x 6
```

```
##   SampleType      n MedianDistance MinDistance MaxDistance      Range
```

##	<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	Feces	4	0.439	0.336	0.532	1.96e- 1
## 2	Freshwater	2	0.392	0.392	0.392	0
## 3	Soil	3	0.379	0.301	0.588	2.87e- 1
## 4	Ocean	3	0.350	0.302	0.477	1.75e- 1
## 5	Skin	3	0.314	0.226	0.598	3.72e- 1
## 6	Tongue	2	0.295	0.295	0.295	5.55e-17
## 7	Sediment (estuary)	3	0.278	0.212	0.407	1.95e- 1
## 8	Freshwater (creek)	3	0.108	0.0449	0.136	9.10e- 2
## 9	Mock	3	0.0405	0.0265	0.0520	2.55e- 2

The table summarises each sample type's distance to its group centroid, providing a clearer view of within-group dispersion than boxplots with few observations.

Freshwater and **Tongue** each contain two samples. Their near-zero distance range is a geometric property of two-sample groups, where both points are equidistant from their midpoint, rather than evidence of low biological variability.

Among groups with at least three samples, **Mock** had the lowest median distance to centroid (0.040), followed by **Freshwater (creek)** (0.108), indicating relatively consistent composition within these groups. **Feces** had the highest median distance (0.439), while **Skin** showed the widest distance range (0.226–0.598), indicating greater within-group heterogeneity. These descriptive patterns are consistent with the significant dispersion test but should be interpreted cautiously because group sizes remain small.

12 Taxonomic composition

Feature-level relative abundances were aggregated to the phylum level and visualised as stacked bar charts. The ten most abundant phyla were shown individually, with all remaining phyla grouped as “Other.”

```
# Agglomerate to phylum level
ps_phylum <- tax_glom(GlobalPatterns_relabund, taxrank = "Phylum")

tax_df <- as.data.frame(tax_table(ps_phylum)) %>%
  rownames_to_column("OTU")

# Determine the 10 most abundant phyla by mean relative abundance.
# IMPORTANT: taxa_sums() names are OTU IDs, not phylum names, so they
# must be resolved via tax_df before comparing against the Phylum column.
phylum_means <- taxa_sums(ps_phylum) / nsamples(ps_phylum)
top10_otu <- names(sort(phylum_means, decreasing = TRUE))[1:10]
top10_phyla <- tax_df$Phylum[match(top10_otu, tax_df$OTU)] %>% as.character()

top10_phyla # sanity check: should be 10 readable phylum names, e.g. "p__Bacteroidetes"
```

```
## [1] "Proteobacteria" "Firmicutes" "Bacteroidetes" "Cyanobacteria"
## [5] "Actinobacteria" "Verrucomicrobia" "Acidobacteria" "Planctomycetes"
## [9] "Fusobacteria" "Tenericutes"
```

```
# Pull out taxonomy + abundance as a long data frame for plotting, since
# this gives explicit control over the "Other" grouping
otu_df <- as.data.frame(otu_table(ps_phylum))
if (!taxa_are_rows(ps_phylum)) otu_df <- t(otu_df)
```

```

otu_df <- otu_df %>%
  as.data.frame() %>%
  rownames_to_column("OTU") %>%
  pivot_longer(-OTU, names_to = "SampleID", values_to = "RelAbundance")

phylum_long <- otu_df %>%
  left_join(tax_df %>% select(OTU, Phylum), by = "OTU") %>%
  left_join(
    sample_data(ps_phylum) %>% as("data.frame") %>%
      rownames_to_column("SampleID") %>%
      select(SampleID, SampleType),
    by = "SampleID"
  ) %>%
  mutate(Phylum = ifelse(Phylum %in% top10_phyla, as.character(Phylum), "Other")) %>%
  group_by(SampleID, SampleType, Phylum) %>%
  summarise(RelAbundance = sum(RelAbundance), .groups = "drop")

# Sanity check: should show 11 distinct values (10 phyla + "Other"),
# not a single dominant category
phylum_long %>% count(Phylum, sort = TRUE)

```

```

## # A tibble: 11 x 2
##   Phylum      n
##   <chr>      <int>
## 1 Acidobacteria    26
## 2 Actinobacteria   26
## 3 Bacteroidetes    26
## 4 Cyanobacteria    26
## 5 Firmicutes       26
## 6 Fusobacteria     26
## 7 Other            26
## 8 Planctomycetes   26
## 9 Proteobacteria   26
## 10 Tenericutes     26
## 11 Verrucomicrobia 26

```

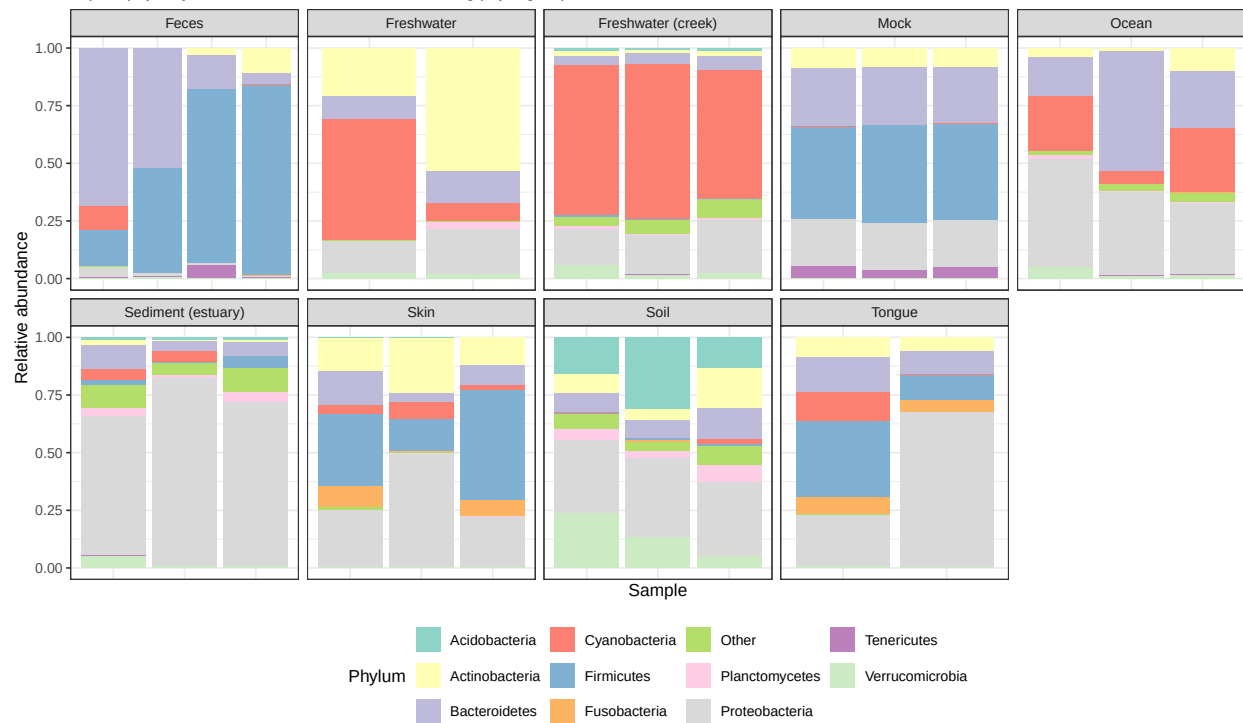
```

ggplot(phylum_long, aes(x = SampleID, y = RelAbundance, fill = Phylum)) +
  geom_col() +
  facet_wrap(~ SampleType, scales = "free_x", nrow = 2) +
  scale_fill_brewer(palette = "Set3") +
  theme_bw() +
  theme(
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    legend.position = "bottom"
  ) +
  labs(
    title = "Taxonomic composition at phylum level",
    subtitle = "Top 10 phyla by mean relative abundance; remaining phyla grouped as \"Other\"",
    x = "Sample",
    y = "Relative abundance",
    fill = "Phylum"
  )

```

Taxonomic composition at phylum level

Top 10 phyla by mean relative abundance; remaining phyla grouped as "Other"



12.0.1 Dominant phyla by sample type

```
phylum_long %>%
  group_by(SampleType, Phylum) %>%
  summarise(MeanRelAbundance = mean(RelAbundance), .groups = "drop") %>%
  group_by(SampleType) %>%
  slice_max(MeanRelAbundance, n = 3) %>%
  arrange(SampleType, desc(MeanRelAbundance))
```

```
## # A tibble: 27 x 3
## # Groups:   SampleType [9]
##   SampleType      Phylum      MeanRelAbundance
##   <fct>          <chr>          <dbl>
## 1 Feces          Firmicutes        0.548
## 2 Feces          Bacteroidetes     0.352
## 3 Feces          Actinobacteria    0.0347
## 4 Freshwater     Actinobacteria    0.371
## 5 Freshwater     Cyanobacteria     0.302
## 6 Freshwater     Proteobacteria    0.167
## 7 Freshwater (creek) Cyanobacteria    0.626
## 8 Freshwater (creek) Proteobacteria    0.186
## 9 Freshwater (creek) Other           0.0597
## 10 Mock          Firmicutes        0.414
## # i 17 more rows
```

The table summarises the three most abundant phyla within each sample type.

Result: Feces samples were dominated by Firmicutes (54.8%) and Bacteroidetes (35.2%). Freshwater creek samples were dominated by Cyanobacteria (62.6%), followed by Proteobacteria (18.6%). Non-creek freshwater samples were more balanced, with Actinobacteria (37.1%), Cyanobacteria (30.2%), and Proteobacteria (16.7%) as the most abundant phyla. Mock communities were dominated by a small set of phyla, including Firmicutes (41.4%). These patterns help explain the observed diversity results: Cyanobacteria dominance is consistent with the high richness but lower Shannon diversity of freshwater creek samples, while the consistent compositions of creek and mock samples align with their lower within-group dispersion.

13 Interpretation and limitations

This analysis applied a standard downstream 16S microbiome workflow: sequencing-depth assessment, conservative filtering, alpha diversity, beta diversity, and phylum-level taxonomic composition.

Overall, sample type was associated with both alpha diversity and community composition. Soil samples showed high richness and evenness, whereas freshwater creek samples combined high richness with uneven taxonomic composition dominated by Cyanobacteria. Sample type was strongly associated with Bray–Curtis community composition, although unequal within-group dispersion means this result should be interpreted cautiously.

13.1 Limitations

- The analysis used preprocessed 16S data and did not reproduce raw-read processing, denoising, chimera removal, or taxonomy assignment.
- Sequencing counts are compositional; relative abundance does not measure absolute microbial abundance.
- The dataset includes heterogeneous environments and small, unequal sample-type groups, so findings are descriptive and exploratory.
- Alpha diversity was calculated from rarefied counts. Rarefaction standardises depth but discards reads, and low-depth samples may underestimate true richness.
- Dispersion summaries for Freshwater and Tongue are not informative because each group contains only two samples.
- No differential-abundance testing was performed.

14 Reproducibility information

This report was generated in R. The session information below records the R version, operating system, and package versions used for the analysis.

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 10 x64 (build 19045)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=en_US.UTF-8  LC_CTYPE=en_US.UTF-8    LC_MONETARY=en_US.UTF-8
## [4] LC_NUMERIC=C            LC_TIME=en_US.UTF-8
```

```

##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] microbiome_1.34.0 phyloseq_1.56.0  janitor_2.2.1    here_1.0.2
## [5] knitr_1.52        rmarkdown_2.32   ape_5.8-1        vegan_2.7-6
## [9] permute_0.9-10    lubridate_1.9.5  forcats_1.0.1    stringr_1.6.0
## [13] dplyr_1.2.1       purrr_1.2.2      readr_2.2.0      tidyr_1.3.2
## [17] tibble_3.3.1      ggplot2_4.0.3    tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] ade4_1.7-24        tidyselect_1.2.1  farver_2.1.2
## [4] Biostrings_2.80.2  S7_0.2.2          fastmap_1.2.0
## [7] digest_0.6.39      timechange_0.4.0  lifecycle_1.0.5
## [10] cluster_2.1.8.2    survival_3.8-6    magrittr_2.0.5
## [13] compiler_4.6.1     rlang_1.3.0       tools_4.6.1
## [16] utf8_1.2.6         igraph_2.3.3      yaml_2.3.12
## [19] data.table_1.18.6.1 labeling_0.4.3     plyr_1.8.9
## [22] RColorBrewer_1.1-3 Rtsne_0.17         withr_3.0.3
## [25] BiocGenerics_0.58.1 grid_4.6.1         stats4_4.6.1
## [28] multtest_2.68.0    biomformat_1.40.0 scales_1.4.0
## [31] iterators_1.0.14   MASS_7.3-65        cli_3.6.6
## [34] crayon_1.5.3       generics_0.1.4     otel_0.2.0
## [37] rstudioapi_0.19.0  reshape2_1.4.5     tzdb_0.5.0
## [40] splines_4.6.1      parallel_4.6.1     XVector_0.52.0
## [43] vctrs_0.7.3        Matrix_1.7-5       jsonlite_2.0.0
## [46] IRanges_2.46.0     hms_1.1.4          S4Vectors_0.50.3
## [49] foreach_1.5.2      glue_1.8.1         codetools_0.2-20
## [52] stringi_1.8.9      gtable_0.3.6       pillar_1.11.1
## [55] htmltools_0.5.9    Seqinfo_1.2.0      R6_2.6.1
## [58] rprojroot_2.1.1    evaluate_1.0.5     lattice_0.22-9
## [61] Biobase_2.72.0     snakecase_0.11.1   Rcpp_1.1.2
## [64] nlme_3.1-169       mgcv_1.9-4         xfun_0.61
## [67] pkgconfig_2.0.3

```